

Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS DELIVERY

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Summary

Healthcare Data Consultant with 10+ years delivering governed analytics platforms and ML solutions across clinical, operational, and revenue-cycle domains. Combines hands-on Python, SQL, Snowflake, AWS, and Azure expertise with leadership in data quality, reconciliation, technical standards, and cross-functional delivery.

Experience

HHaEXchange

Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox

HEALTHCARE DATA INTEGRATION CONSULTANT — CONTRACT VIA VEERSA TECHNOLOGIES

Oct. 2025 - Jun. 2026

Led healthcare source analysis, data-quality investigation and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains, centralizing 37% of post-source transformation logic in shared code.
- Created or materially redesigned 30 SQL/dbt data-quality tests and designed macros to automate null, duplicate, referential-integrity, business-rule, lifecycle, and source-to-target checks.
- Profiled source schemas and converted ambiguous business rules into documented mappings, acceptance criteria, and onboarding standards, plus 12 data-quality runbooks that standardized investigation and onboarding practices.

Adonis

AWS: S3, Snowflake, SQLMesh, SQL, Python

HEALTHCARE REVENUE CYCLE DATA CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS

Mar. 2025 - Jul. 2025

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models, supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated validation across 54 SQLMesh models, including exact row-count audits, uniqueness checks, and not-null checks; achieved a 100% pass rate across 117+ configured audit applications at final delivery, including exact row-count reconciliation for 40 models, with no unresolved discrepancies.
- Diagnosed undocumented source-key versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models in one day, eliminating duplicate-key failures and directly unblocking final delivery.
- Authored implementation playbooks, plus internal handoff documentation and a QA runbook that established a repeatable healthcare-data validation workflow for an engineering team new to the domain.

GrowData Analytics

Ruby, RubyLLM, Tesseract OCR, GPT-4o mini, Claude Opus/Sonnet, RuboCop, RSpec

INDEPENDENT DATA AUTOMATION & ANALYTICS CONSULTANT

Sep. 2024 - Oct. 2025

Built a document-to-data product that converted semi-structured invoices into validated, auditable transaction records.

- Built and validated a pre-production Ruby, OCR, and LLM invoice-automation MVP within 11 months, achieving 95% field-level accuracy across a stratified 500-invoice test suite (95% exact CI: 92.67%-96.79%).
- Instrumented the MVP with RSpec validation, audit logging, retry controls, error classification, and confidence-based routing, demonstrating 88% straight-through processing, 12% manual review, and under 1% complete system failure in testing.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Delivered healthcare analytics, recurring reporting and data-quality investigations while improving the governed data foundation used by the BI team.

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets.
- Integrated Jupyter-based reporting and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring and executive reporting.

Merative

Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Led data validation, compliance and migration assurance for a regulated healthcare platform modernization.

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition-level reconciliation, schema compatibility, null handling, duplicates, field-level comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity with zero unresolved discrepancies.
- Led HIPAA-compliant de-identification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.

IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Delivered client analytics and large-scale migration validation for healthcare administrative data.

- Developed an MVP surgical-scheduling optimizer and a CNN/RNN case-card prediction proof of concept, automating case-card generation and resource allocation across operating rooms, time slots, and personnel.
- Engineered a reusable Spark, Hive, and Impala analytics framework powering 50+ downstream prior-authorization lag-time and approval-bottleneck reports, with daily operational execution and monthly or quarterly regulatory and financial snapshots.

SENIOR DATA ARCHITECT

Dec. 2020 - Nov. 2021

Led clinical-data integration and unstructured-text analysis at enterprise scale.

- Architected Spark and Hadoop pipelines that integrated and normalized more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of standardized measures and registries.
- Replaced static mappings with executable, metadata-driven instruction sets adopted across approximately 200 source integrations, standardizing EDI batch schemas while enabling modular integration templates, automated Python/Jupyter lineage tracing, and end-to-end validation gates.

DATA ARCHITECT

Nov. 2017 - Dec. 2020

Owned client-facing integration, discrepancy resolution and data-quality improvement across multi-organization healthcare networks.

- Designed standardized SQL validation frameworks and client-review gates across approximately 100 client integrations, reducing Jira-tracked validation rework hours by 25% and mapping rework hours by 45% across initial delivery and subsequent corrections.
- Led architecture and delivery of a multi-organization healthcare integration spanning 70+ heterogeneous EHR sources and isolated organizational schemas; implemented provider-scoped patient filtering to enforce data-sharing rules, and coordinated with engineering teams.

SOFTWARE DEVELOPER, DATA PLATFORM

Sep. 2016 - Nov. 2017

Built internal data-curation and knowledge-transfer tools that reduced costs and onboarding effort.

- Engineered an in-house HL7 v2/v3 parsing pipeline in Pig, supporting approximately 30 internal users, replacing the Mirth-based workflow, and eliminating \$30,000 in annual vendor costs.
- Built a Python, Ruby, and SQL extraction tool that reduced total client-integration effort by 28%, measured through eliminated story points and corroborated by lower average engineering hours per implementation.

DATA SCIENTIST

Nov. 2015 - Sep. 2016

Coordinated client integration delivery, contractor work and process automation.

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENCE CONSULTANT — CONTRACT VIA TEKSYSTEMS

Jun. 2015 - Nov. 2015

Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.

Other Experience

SOFTWARE ENGINEER, **AmTrust Financial Services**, *Mar. 2014 - Jun. 2015.*

COLOR SCIENTIST, **PPG Industries**, *Apr. 2012 - Jun. 2013.*

ADJUNCT PROFESSOR, **Multiple Institutions**, *Aug. 2010 - May. 2019.*

Volunteer Experience

HSEA

ETL BRIGADE

Feb. 2025 - May. 2025

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll

INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

Professional Recognition

HDM, SPECIAL RECOGNITION (2025), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (2023), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (2022), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (2018 - 2022), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (2018), Recognized for coordinating internal teams to deliver solutions on time.

IBM, EMINENCE AND EXCELLENCE AWARD (2017), Recognized for developing a new data curation tool.

PPG, COLOR LAB AUTOMATION AWARD (2012), Recognized for expanding data collection and analysis reporting.

Speaking Engagements

AI Hackathon 2020: Surgery Scheduling Optimization

"An algorithm uses past utilization and current waitlist info to predict surgery duration.

Dec. 2020

IBM

REACH Initiative: IBM Watson Care Insights

Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.

2019

IBM

Content & Data Analytics Learning Exchange

Introduction of implementation of machine learning algorithms using real-world health data analytics.

2018

IBM

2018 Innovation Research Interchange Fall Networks Conference

Using NLP to extract insight from unstructured data and transform it from a data lake.

Sep. 2018

IRI

Cross Team Training Series

A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.

2016

EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets

Reproducible Postgres load of Tuva seed datasets with robust validation.

Aug. 2025

HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction ^a

Improvement on traditional fixed ratio methods for total procedure time estimation.

Dec. 2020

IBM

Process and Requirements for CCD Data Integration

Processes and requirements for overcoming the many difficulties with CCD data curation.

Oct. 2017

IBM

AI Paint Formula Prediction from Spectroscopic Measurements ^b

Naive bayesian color and effects matching algorithm of complex automotive paint formula.

May. 2015

SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching

Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.

Feb. 2013

USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy ^c

Spectroscopic and Photometric analysis of two open clusters in the galactic plane.

Dec. 2010

OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.

May. 2008

VIZIER

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107

New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).

Feb. 2008

PASP

A Novel Method for Low-Toxic Photo-Etch Printing ^d

Application of aquatint etching techniques to photo-etch procedures.

May. 2003

MAPC

Education & Credentials

2022 **IBM Data Science Professional Certificate** ^a

IBM

2015 **Certificate of Specialization** Data Science ^b

Johns Hopkins University

2010 **Master of Science** Astrophysics ^c

University of Cincinnati

2004 **Bachelor of Fine Arts** Printmaking ^d

Ohio University